CS 771A: Intro to Machine Learning, IIT Kanpur				**Quiz I** (29 Jan 2026)
Name				**20 marks**
Roll No		Dept.		Page **1** of **2**

Instructions:

1. This question paper contains 1 page (2 sides of paper). Please verify.
2. Write your name, roll number, department above in **block letters neatly with ink**.
3. Write your final answers neatly **with a blue/black pen**. Pencil marks may get smudged.
4. Don't overwrite/scratch answers especially in MCQ – such cases may get straight 0 marks.
5. Do not rush to fill in answers. You have enough time to solve this quiz.

Q1 (Mix n Match) For each function, select all properties satisfied by that function. A property should be selected only if the function satisfies it for all $x \in \mathbb{R}$. If a function satisfies no property, select NOTA. *Hint: Affine functions look like $f(x) = ax + b$ for some const $a, b \in \mathbb{R}$.* (**5 marks**)

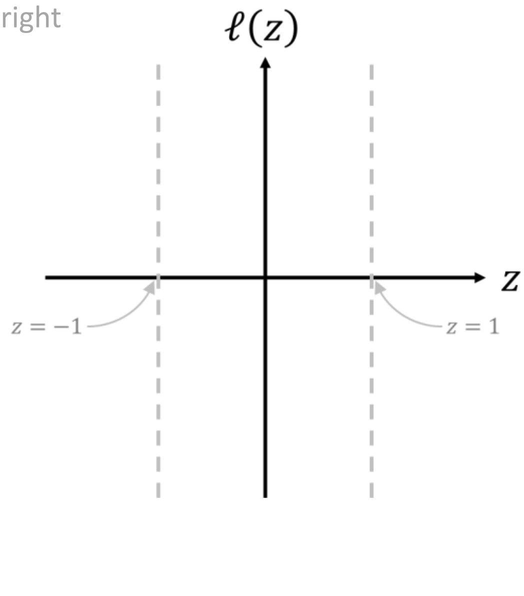
	Function	Continuous?	Differentiable?	Convex?	Affine?	NOTA
a.	$f(x) = \sin(x)$	◯	◯	◯	◯	◯
b.	$g(x) = \max\{x, 0\} - \max\{-x, 0\}$	◯	◯	◯	◯	◯
c.	$h(x) = \operatorname{sign}(x), \operatorname{sign}(0) \stackrel{\text{def}}{=} 0$	◯	◯	◯	◯	◯
d.	$p(x) = \exp(x^2)$	◯	◯	◯	◯	◯
e.	$q(x) = 1$ (constant function)	◯	◯	◯	◯	◯

Q2 (Least Squares SVM?) Melba wants to apply least squares to binary classification due to its simplicity. Given N data points $(\mathbf{x}^i, y^i), i = 1, \dots, N$ with $\mathbf{x}^i \in \mathbb{R}^d, y^i \in \{-1, +1\}$, Melba tries to solve $\min_{\mathbf{w} \in \mathbb{R}^d} \frac{1}{2} \cdot \|\mathbf{w}\|_2^2 + \sum_{i \in [N]} \left(\mathbf{w}^\top \mathbf{x}^i - y^i\right)^2$ (hidden bias). Melbo says that if the labels are either -1 or $+1$, then this is equivalent to solving $\min_{\mathbf{w} \in \mathbb{R}^d} \frac{1}{2} \cdot \|\mathbf{w}\|_2^2 + \sum_{i \in [N]} \ell\left(z^i\right)$ where $z^i \stackrel{\text{def}}{=} y^i \cdot \mathbf{w}^\top \mathbf{x}^i$ i.e., ℓ has a form similar to the hinge loss. In the space provided, show that Melbo is right. Write the equation of ℓ, sketch its graph approximately, and show brief derivation. (**2+1+2 = 5 marks**)

Write equation of loss function ℓ here

$\ell(z) =$

Give brief justification and approximately sketch the graph of ℓ on the right



Q3 (Melbo's Mistake) Melbo wants to simplify the SVM by maximizing the mean distance to the hyperplane and not the smallest distance. For N points $(\mathbf{x}^i, y^i), i \in [N]$, $\mathbf{x}^i \in \mathbb{R}^d, y^i \in \{-1, +1\}$, Melbo wants to solve $\max_{\mathbf{w}\in\mathbb{R}^d} \left\{\sum_{i\in[N]} \frac{|\mathbf{w}^\top \mathbf{x}^i|}{\|\mathbf{w}\|_2}\right\}$ s. t. $y^i \cdot \mathbf{w}^\top x^i \geq 0$ for all i (no bias term). Simplifying as we did in class gives $\max_{\mathbf{w}\in\mathbb{R}^d} \left\{\sum_{i\in[N]} \frac{y^i \cdot \mathbf{w}^\top \mathbf{x}^i}{\|\mathbf{w}\|_2}\right\}$ s. t. $y^i \cdot \mathbf{w}^\top x^i \geq 0$ for all i. Melbo argues that since the constraints $y^i \cdot \mathbf{w}^\top \mathbf{x}^i \geq 0$ have already been used to simplify, we can remove them and just solve $\max_{\mathbf{w}\in\mathbb{R}^d} \left\{\sum_{i\in[N]} \frac{y^i \cdot \mathbf{w}^\top \mathbf{x}^i}{\|\mathbf{w}\|_2}\right\}$ without any constraints. Show that Melbo is incorrect. **(6+2+2 = 10 marks)**

(a) Solve $\arg\max_{\mathbf{w}\in\mathbb{R}^d} \left\{\sum_{i\in[N]} \frac{y^i \cdot \mathbf{w}^\top \mathbf{x}^i}{\|\mathbf{w}\|_2}\right\}$. Assume $\sum_{i\in[N]} y^i \cdot \mathbf{x}^i \neq \mathbf{0}$. Multiple solutions exist (easy to see that if $\mathbf{w}$ is a solution, so is $c \cdot \mathbf{w}$ for $c > 0$). Thus, please give a unit norm solution i.e. $\|\mathbf{w}\|_2 = 1$.

Give brief derivation for the unit norm solution to the problem

(b) Melba created a large dataset $N = 10^{101}, d = 2$ where Melbo's method fails. There are an equal number of points with $+1$ and -1 labels. One +ve point has feature $\mathbf{x}^i = (1, -1)$). All other +ve points have feature $\mathbf{x}^i = (1,1)$. One -ve point has $\mathbf{x}^i = (-1,1)$. All other -ve points have $\mathbf{x}^i = (-1, -1)$. Note that $\mathbf{x}^i \in \mathbb{R}^2, \mathbf{w} \in \mathbb{R}^2$.There is no bias term in the model (not even a hidden one).

(b1) What unit-norm classifier $\mathbf{w} \in \mathbb{R}^2$ would Melbo's method learn on Melba's dataset? (assume $10^{-100} \approx 0$)

$\mathbf{w} = [\_\_\_\_, \_\_\_\_]$

(b2) What classifier $\mathbf{w} \in \mathbb{R}^2$ would hard SVM (no slacks) learn on this dataset? This vector need not be unit norm.

$\mathbf{w} = [\_\_\_\_, \_\_\_\_]$